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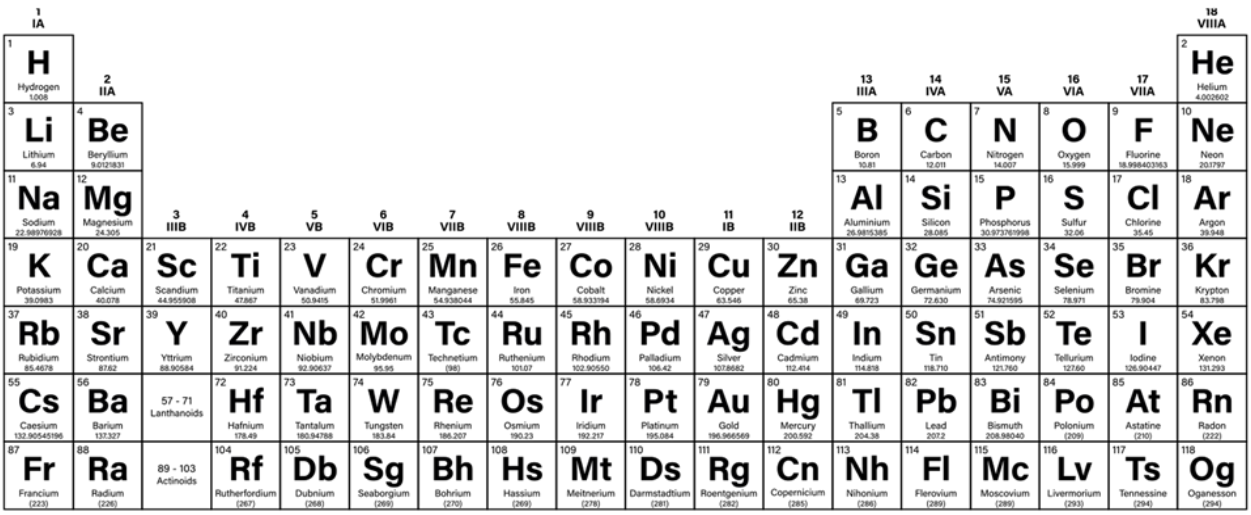
Instructions for CM2021 Mid-term Test

1. This test paper contains forty (40) multiple-choice questions

2. Answer ALL questions (1 mark each)

3. Time allowed: 1 hour

4. Do not turn over the test paper until told to do so



1) Which of the following is not a coordination compound?

a. Urea
b. Haemoglobin
c. Chlorophyll II
d. Cobalt(III) ammines

2) The number of chlorine atoms acting as ligands in the complex $[Co(en)_2(H_2)Cl]Cl_2$ is:

a. 1
b. 3
c. 2
d. None of these

3) Consider the complex $[Co(NH_3)_5CO_3]ClO_4$, in which coordination number, oxidation number, and the number of d-electrons on the metal are respectively?

a. 6, 3, 6
b. 6, 2, 7
c. 5, 3, 6
d. 5, 3, 7

4) Which of the following compound shows both linkage and ionization isomerism?

a. $[Co(NH_3)_5NO_2](NO_2)$
b. $[Co(NO_2)(py)_2(NH_3)_2]NO_2$
c. $[Co(NH_3)_6]Cl_2$
d. $[Co(NO_2)Br(NH_3)_4]Cl$

5) The complex $[Co(NH_3)_5Br]SO_4$ will give white precipitate with:

a. $PbCl_2$
b. $AgNO_3$
c. KI
d. None

6) The name of the coordination compound $[Cu(NH_3)_4](NO_3)_2$ is:

a. cuprammonium nitrate
b. tetraammine copper(II) dinitrate
c. tetraamminecopper(II) nitrate
d. tetraammine copper(III) dinitrate

7) The name of $[Fe(CN)_2(CH_3NH_2)_2]$ is:

a. dicyanobis(methylamine)iron(II)
b. dicyanodimethylamineiron(II)
c. dicyanodimethylamineiron(IV)
d. dimethylaminedicyanoiron(II)

8) The name of $[Pt(NH_3)_4Cl_2][PtCl_4]$ is:

a. tetraamminedichloroplatinum(IV) tetrachloroplatinate(II)
b. dichlorotetraammineplatinum(IV) tetrachloroplatinate(II)
c. tetrachloroplatinum(II) tetraammineplatinate(IV)
d. tetrachloroplatinum(II) dichlorotetraammineplatinate(IV)

9) Which of the following coordination complexes is most stable?

a. $[Ni(NH_3)_6]^{2+}$ b. $NiCl_2$

c. $[Ni(H_2O)_6]^{2+}$ d. $[NiEDTA]^{2-}$

10) SCN^- has two possible donor atoms. It is a/an

a. monodentate ligand b. bidentate ligand

c. tridentate ligand d. tetradentate ligand

11) Which of the following gives the most number of isomers?

a. $[Co(NH_3)_4Cl_2]$ b. $Ni(en)(NH_3)_4]^{2+}$

c. $[Ni(C_2O_4)(en)_2]$ d. $[Cr(SCN)_2(NH_3)_4]^{4+}$

12) $[Co(NH_3)_4(NO_2)_2]Cl$ exhibits:

a. ionization, geometrical and optical isomerisms

b. linkage, geometrical and optical isomerisms

c. linkage, ionization and optical isomerisms

d. linkage, geometrical and ionization isomerisms

13) The total number of possible isomers for the square planar $[PtCl(NO_2)(NO_3)(SCN)]^{2-}$ is:

a. 12 b. 8

c. 16 d. 24

14) Optical isomers have:

a. same physical properties b. same molecular formula

c. difference in action towards plane polarized light

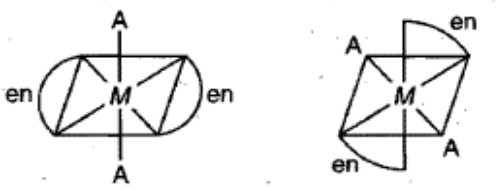
d. all of the above

15) Which of the following compound shows optical isomerism?

a. $[Co(CN)_6]^{3-}$ b. $[Cr(C_2O_4)_3]^{3-}$

b. $[ZnCl_4]^{2-}$ d. $[Cu(NH_3)_4]^{2+}$

16) Two complexes given below are:



a. geometrical isomers b. position isomers

c. optical isomers d. identical

17) Which of the following will have two stereoisomeric forms?
i. $[Cr(NO_3)_3(NH_3)_3]$
ii. $K_3[Fe(C_2O_4)_3]$
iii. $[CoCl_2(en)_2]^+$
iv. $[CoBrCl(ox)_2]^{3-}$

a. i only
b. i and ii
c. iii and iv
d. all of these

18) The octahedral complex $CoSO_4Cl_5NH_3$ exists in two isomeric forms X and Y. Isomers X reacts with $AgNO_3$ to give a white precipitate, but does not react with $BaCl_2$. Isomer Y gives white precipitate with $BaCl_2$ but does not react with $AgNO_3$. Isomers X and Y are:

a. ionization isomers
b. linkage isomers
c. coordinating isomers
d. solvent isomers

19) Which of the following does not have optical isomer?

a. $[Co(NH_3)_3Cl_3]$
b. $[Co(en)_3]Cl_3$
b. $[Co(en)_2Cl_2]Cl$
d. $[Co(en)(NH_3)_2Cl_2]Cl$

20) Number of geometrical isomers shown by the octahedral complex of the type $[MX_2Y_4]$ is

a. 4
b. 6
c. 8
d. 2

21) Transition metals form most stable complex ions as:

a. they exhibit multiple oxidation states
b. they have large charge to radius ratio
c. vacant orbitals to accommodate electrons are present
d. all of these

22) Which of the following metal atomic orbital is considered in crystal field theory?

a. s
b. p
c. d
d. f

23) The more stable complex among the following is:

a. $[Fe(H_2O)_6]^{2+}$
b. $[Mn(H_2O)_6]^{2+}$
c. $[Fe(H_2O)_6]^{3+}$
d. None of these

24) Stability of a metal complex depends upon:

a. charge on the central metal ion
b. nature of the ligand
c. chelation
d. all of the above

25) The stability constant of the complexes formed by a metal ion M^{2+} with NH_3, CN^-, H_2O, and en are of the order of 10^{11}, 10^{27}, 10^{15}, and 10^{18}, respectively. Which of the following is true:

a. CN^- is the strongest ligand
b. en is the strongest ligand
c. these values cannot predict the strength of the ligand
d. all the ligands are equally strong

26) The degeneracy of d-orbitals is lost under:

i. strong field ligand
ii. weak field ligand
iii. mixed field ligand
iv. chelated ligand field

a. i, ii and iv
b. i and ii
c. i, ii, iii and iv
d. i, ii and iii

27) Choose the incorrect order of stability:

a. $[Cu(NH_3)_4]^{2+} < [Cu(en)_2]^{2+} < [Cu(trien)]^{2+}$
b. $[Fe(H_2O)_6]^{3+} < [Fe(NO_2)_6]^{3-} < (Fe(NH_3)_6]^{3+}$
c. $[Fe(H_2O)_6]^{2+} < [Fe(H_2O)_6]^{3+} < [Co(NH_3)_6]^{3+}$
d. $[Co(H_2O)_6]^{3+} < [Rh(H_2O)_6]^{2+} < [Ir(H_2O)_6]^{3+}$

28 The complexes formed by Cu^{2+} ion are more stable than those formed by Cd^{2+} ion because:

a. the value of the stability constant for Cd^{2+} complexes is greater than that of Cu^{2+} ions
b. the charge density on Cu^{2+} is greater than that of Cd^{2+} ion
c. the ionic radius of Cu^{2+} ion is more than that of Cd^{2+} ion
d. Cu^{2+} ion forms chelate compounds

29) The most stable complex species is:

a. $[Fe(C_2O_4)_3]^{3-}$
b. $[Fe(CN)_6]^{3-}$
c. $[Fe(CN)_6]^{4-}$
d. $[Fe(H_2O)_6]^{3+}$

30) In $Fe(CO)_5$, the Fe-C bond possesses:

a. π character only
b. σ character only
c. ionic character only
d. both π and σ character

31) In the complex ion ML_6^{n+} ,M^{n+} has five d-electrons and L is weak field ligand. According to crystal field theory, how many unpaired electrons are there in this complex ion?

a. 0
b. 5
c. 2
d. 3

32) The energies of d_{xy} and d_z^2 orbitals in octahedral and tetrahedral transition metal complexes are such that:

a. $E(d_{xy}) > E(d_z^2)$ in both tetrahedral and octahedral complexes
b. $E(d_{xy}) < E(d_z^2)$ in both tetrahedral and octahedral complexes
c. $E(d_{xy}) > E(d_z^2)$ in tetrahedral but $E(d_{xy}) < E(d_z^2)$ in octahedral complexes
d. $E(d_{xy}) < E(d_z^2)$ in tetrahedral but $E(d_{xy}) < E(d_z^2)$ in octahedral complexes

33) What is the electronic arrangement of metal atom/ion in octahedral complex with d^4 configuration, if Δ_o < pairing energy

a. $t_{2g}^4 e_g^0$
b. $e_g^4 t_{2g}^0$
c. $t_{2g}^3 e_g^1$
d. $e_g^2 t_{2g}^2$

34) The ratio of CFSE for $[NiCl_6]^{4-}$ and $[NiCl_4]^{2-}$ will be nearly:

a. 3/2
b. 4/9
c. 9/4
d. 27/8

35) The d orbitals that are stabilized in an octahedral magnetic field are:

a. d_{xy} and d_{z2}
b. d_{x2-y2} and d_{z2}
c. d_{xy}, d_{xz} and d_{yx}
d. d_{z2} only

36) The magnitude of crystal field stabilization energy in tetrahedral complexes is considerably less than in the octahedral field because:

a. there are only four ligands instead of six
b. the direction of the orbitals does not coincide with the direction of the ligands
c. Both points (a) and (b) are correct
d. Both points (a) and (b) are wrong

37) Which of the following on CO ligand is true?

a. σ-donor, π-acceptor
b. σ-donor, π-donor
c. σ-acceptor, π-acceptor
d. σ-acceptor, π-acceptor

38) Cisplatin compounds are used in the treatment of:

a. malaria
b. cancer
c. AIDS
d. yellow fever

39) In the spectrochemical series, which ligand produces strongest field?

a. Cl^-
b. H_2O
c. NO_2
d. CO

40) Which effect/theory could account for the presence of two different M-L bond lengths in $[Cu(H_2O)_6]^{2+}$?

a. Jahn-Teller effect
b. Crystal Field Theory
c. Ligand Field Theory
d. Valence Bond Theory